Reg. No. [ ] [17MT45]

FOURTH SEMESTER B.Tech EXAMINATION, JUNE – 2019

IRON MAKING

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least **TWO** questions from each part.
2. Missing data may be suitably assumed.

PART - A

1.	**a.**	What are the major developments observed in recent era for production of Iron?	**07**
	b.	Write any seven essential characteristics of iron ore.	**07**
2.	**a.**	Explain sintering process and mention the types of sinters. Draw temperature prevailing in a sinter bed after ignition.	**08**
	b.	Explain briquetting and nodulizing process.	**06**
3.	**a.**	Explain mechanism of ball formation in pelletization process.	**08**
	b.	With neat sketch explain straight travelling grate process.	**06**
4.	**a.**	Draw schematic sketch of blast furnace indicating different parts.	**07**
	b.	Write about bell charging system with neat diagram.	**07**

PART - B

5.	**a.**	Write down the idealized scheme of temperature distribution of gas and solids along the height of the blast furnace and the chemical reactions occurring in the main temperature zones.	**08**
	b.	Write about the endothermic reactions occurring in the lower zone of blast furnace.	**06**
6.	**a.**	A blast furnace makes hot metal containing 3.6 wt. % C, 1.4 % Si, the remainder being Fe. Calculate the weight of ore used for following conditions, **i.** The ore contains 8.5 % Fe_2O_3, the remainder being 15% gangue of SiO_2 and Al_2O_3. **ii.** The coke contains 85 % fixed carbon and 15 % ash.	**06**
	b.	Briefly explain tuyere flame temperature and raceway adiabatic flame temperature.	**08**
7.	**a.**	How do you clean the blast furnace gas? Explain in detail.	**07**
	b.	Briefly explain about channeling and scaffolding in blast furnace.	**07**
8.	**a.**	Write about smelting reduction process.	**07**
	b.	What is direct reduction processes? Describe fluidised bed process.	**07**

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